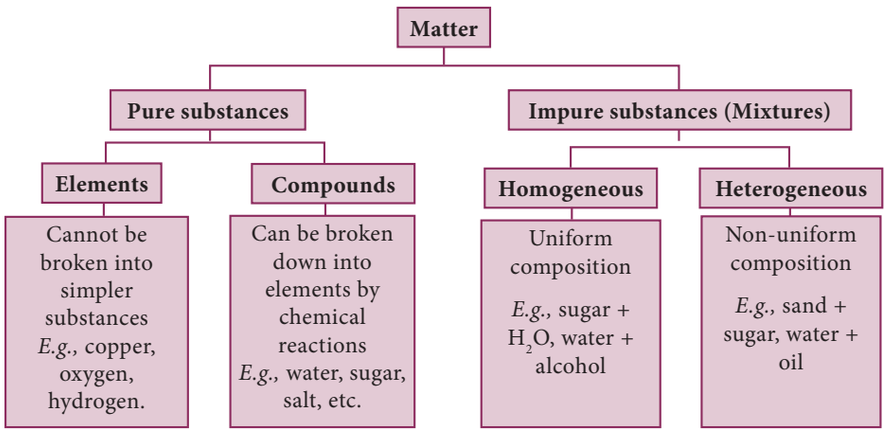


H, He, Li……. 118 elements

Building block of all materials

Activity 1

1. Is air a pure substance or Mixture? Justify
2. You must have seen brass statues in museums and places of worship. Brass is an alloy made up of approx. 30% zinc and 70% copper. Is Brass a pure substance or a mixture or compound?

10.1.1 Elements

Most of you may be interested in music, and some of you may know how it is composed. Music is the combination of a few basic musical notes i.e., Sa, Re, Ga,…Thus, the building blocks of music are the musical notes.

Sa, Re, Ga, Ma, Pa...

Building blocks of music

Likewise , all substances on earth are made up of certain simple substances called elements. Plants, cats, apples, rocks, cars and even our bodies contain elements. Thus, elements are the building block of all materials.



In the modern periodic table there are 118 elements known to us, 92 of which are naturally occurring while the remaining 26 have been artificially created. But from these 118 elements, crores of compounds are formed- some naturally occurring and some artificial. Isn't that amazing?

Robert boyle used the name element for any substance that cannot be broken down further, into a simpler substance. This definition can be extended to include the fact that each element is made up of only one kind of atom. For example, aluminium is an element which is made up of only aluminium atoms. It is not possible to obtain a simpler substance chemically from the aluminium atoms. You can only make more complicated substances from it, such as aluminium oxide, aluminium nitrate and aluminum sulphate.

Atom: The smallest unit of an element which may or may not have an independent existence, but always takes part in a chemical reaction is called atom.

Molecules: The smallest unit of a pure substance, which always exists independently and can retain physical and chemical properties of that substance is called a molecule.

Examples:

Hydrogen molecule consists of two hydrogen atoms (H_2)

Water molecule consists of two hydrogen atoms (H_2) and one oxygen atom (O).

All elements can be classified according to various properties. A simple way to do this is to classify them as metals, non metals and metalloids.

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